

JASON MATTHEW WIDJAJA

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EDUCATION

Northeastern University

Boston, MA

Master of Science in Robotics, Mechanical Engineering Concentration | GPA: 3.8

May 2026

Bachelor of Science in Mechanical Engineering, Robotics Minor | GPA: 3.9

May 2025

Activities: Generate, ASME, Dragon Boat, IM Sports, NTU Singapore Study Abroad, NEU Seattle Aerospace Program

EXPERIENCE

Berkshire Grey

July 2025 - Dec 2025

Mechanical Engineer Co-Op

Bedford, MA

- Drove design reviews and custom part release for electromechanical assemblies, pushing ECO release (Arena PLM) through internal CCB reviews and coordinating vendors/on-site engineers for on-time multi-site installations
- Delivered a linear-actuator test rig demonstrating 1" centerline tracking accuracy to support wag controls tuning
- Converted a machined PLI body to sheet-metal with CNC ribs, reducing part cost by 65% and lead time by >2 weeks
- Deployed a recirculation chute that reduced SQA cycling times by 30% and reduced costs of up to \$10K+
- Piloted system-level improvements to conveyor drivetrain through deterministic jack-screw tensioning, COTS component selection to reduce cost, and DFA-driven consolidation of custom drivetrain cover

Draper

July 2024 - Dec 2024

Electro-Mechanical Engineer Co-Op

Cambridge, MA

- Enhanced IFOG (interferometric fiber-optic gyroscope) integration hardware and documentation, cutting assembly processing time by 1 hour per unit across programs and confirming results through metrology-based studies
- Owned drawing release for 30+ part/assembly packages to ASME Y14.5 standards with GD&T, managing PDM configuration control and vendor communication to ensure compliant fabrication
- Built a MATLAB-LTspice workflow to sweep circuit parameters and identify voltage-spikes, achieving target frequencies

Amazon Robotics

July 2023 - Dec 2023

R&D Hardware Engineer Co-Op

Westborough, MA

- Designed and built two early-stage pan-tilt prototypes for an OTS industrial camera in the Innovation Lab, using hand calcs/FEA to verify loads and programming two stepper motors for full user-controlled pan/tilt motion
- Implemented wireless pan-tilt control in Python via SSH, reducing test costs of over \$15K+ per identification system
- Engineered a camera focus mechanism, improving barcode-label calibration and identification (patent pending)
- Led fixture development for overhead Computer Vision systems and presented work to senior leadership

PROJECTS

Bourbot | Senior Capstone

Jan 2025 - Apr 2025

- Designed a semi-automated tele-op robot to roll, lift, and rotate 550-lb whiskey barrels. Developed a ball-screw driven lifting mechanism. Verified stresses/deflection with FEA and MATLAB calculations and demonstrated live operation.

Generate Product Development | Northeastern University

Sep 2023 - Dec 2024

- **Fitolux — ME Hardware Lead (Fall 2024):** Led a team of five MEs to develop a portable/countertop powder-dispensing device. Drove requirements, design reviews, and rapid prototyping for mechanical systems.
- **C-STAR — ME Hardware Engineer (Spring 2024):** Designed the sounding-tool subsystem for an autonomous drive unit performing concrete delamination detection. Executed benchtop and field tests and iterated for robustness.
- **WaveWise — ME Hardware Engineer (Fall 2023):** Built an IP-65 rated depth-adjustment mechanism for a dynamic vertical kelp-farm prototype and validated in an aquatic environment.

Brailleforge | MakeMIT Hardwarethon

Feb 2023

- Placed **3rd** in overall competition. Led the development of a 2-axis gantry that converts user input text to braille with limited materials in under 15 hours, resulting in more than a 50% reduction in cost compared to existing braille printers.

NASA Big Idea Challenge 2022 | Silicon Synapse Lab

Sep 2021 - Dec 2022

- Awarded \$170K+ and received **The Artemis Award**. Co-developed an 11-joint modular snake robot for extreme environments. Designed power-electronics enclosures, executed locomotion tests, and presented at the Big Idea Forum.

SKILLS AND INTERESTS

Skills: CAD And Design - SOLIDWORKS (CSWP), Onshape, Creo | 3D Printing - FDM, SLA | Machining - Mill, Lathe | Hand/Power Tools | Sheet Metal/Weldments | FEA | GD&T | Aerial Lift | Jira | Arena PLM

Electronics And Programming: Arduino | RPi | Soldering | LTSpice | Python | MATLAB | Simulink | Simscape

Interests: Fitness, Basketball, Baseball, Ping Pong, Outdoors, Cooking, Traveling, F1, Robotics